

Question Paper

Exam Date & Time: 26-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026

SUBJECT: PHY 1071/PHY_1071 ENGINEERING PHYSICS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Derive the relation between different Einstein Coefficients at thermal equilibrium. (5)
Discuss the nature of emission for $hf < KT$.

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) A quantum simple harmonic oscillator consists of an electron bounded by a restoring force proportional to its displacement relative to an equilibrium position. The photon of 550 nm is used to excite the oscillator to its first excited state from the ground state. Calculate the proportionality constant and energy of the first excited state. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 1C) State whether the given statement is true or false and justify your answer. 'At room temperature the probability of finding an electron at the Fermi level is 0.5', (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2A) With the help of schematic diagram describe how the Davisson-Germer experiment confirmed the wave nature of electrons. (5)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2B) A 30 eV electron and proton are incident on a square barrier of height 40 eV. What is the probability that the electron and proton will tunnel through the barrier if its width is 0.10 nm. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2C) State whether the wavefunction $\psi = A e^{Cx}$ for $x \leq 0$ is physically acceptable or not in the finite potential well with finite height 'U' and length 'L'. Justify your answer. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3A) Derive an expression for density-of-states. (5)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3B) The frequency of photon that causes $v = 0$ to $v = 1$ transition in the CO molecule is 6.42×10^{13} Hz. (3)
Ignore any changes in the rotational energy.

- (a) Calculate the force constant k for this molecule.
 (b) What is the maximum classical amplitude of vibration for this molecule in the $v = 0$ vibrational state

use $\mu = 1.14 \times 10^{-26} \text{ kg}$

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3C) The Fermi level in silver is 5.5 eV at zero kelvin. Calculate the number of free electrons per unit volume. (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4A) What are the factors on which the electric field at some point P near a dipole in a dielectric material depends? Obtain a relationship between the polarizability and the relative permittivity of dielectric materials. (5)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4B) The atomic weight of Si is 28 and has a density of 2.33 g cm^{-3} and each atom has polarizability of $2 \times 10^{-40} \text{ F m}^2$. Calculate the ratio of internal field to external field in the case of Si. Given $\epsilon = 8.854 \times 10^{-12} \text{ F/m}$. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) State whether the given statement is true or false and justify your answer. "The density of states in a bulk material can be obtained using quantum free electron theory because the energy levels of electrons are treated as continuous." (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5A) What are biomaterials? Explain its various biomedical applications with suitable examples. (5)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5B) Explain how nanotechnology is applied in microelectronics, optoelectronics and medical applications. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) State whether the given statement is true or false and justify your answer. "The electrical resistivity of metallic glasses is high and remains almost constant with temperature." (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

-----End-----

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026

PHY_1071:ENGINEERING PHYSICS

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	5
3	3	CO 3	22
4	4	CO 4	8
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	PHY_1071:ENGINEERING PHYSICS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	25(50%)
3	Application	9	23(46%)
4	Analysis	1	2(4%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)